PROJECT GUIDANCE

Workshop Guide

From idea to MVP in 4 weeks.

A step-by-step guide for every pod — brainstorming to Blueprint Lock to hackathon-ready deployment on 8th May 2026.

For Students

Use this as your project handbook

For Instructors

Follow section by section in the room

1 Select Project
2 Form Pod
3 Problem Statement
4 AI Co-Founder
5 Tool Stack + Dataset
6 Architecture
7 RACI + Gantt
8 MVP Plan
9 Definition of Done
10 Demo Script
11 Blueprint Lock
12 Build & Ship

AIForEveryone.ai × InTime IT Services | Haldia Institute of Technology | 3rd Year Batch 2026

The Complete Roadmap — From Day 1 to Hackathon Stage

PHASE 1 — FOUNDATION (Complete on Day 1)

- 01 Select Project
- 02 Form Pod
- 03 Problem Statement
- 04 AI Co-Founder
- 05 Tools + Dataset
- 06 Risk Register

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- 07 Architecture
- 08 RACI Matrix
- 09 Gantt Chart
- 10 MVP Plan
- 11 Definition of Done
- 12 Demo Script

✓ Outcome: Problem defined · AI Co-Founder active · Tools chosen · Data loaded · Risks logged

PHASE 2 — BLUEPRINT (Complete on Day 2)

✓ Outcome: Blueprint locked · Ready to execute · Demo scripted

01

Step 1

Select Your Project

Choose a problem worth solving. Everything downstream depends on this decision.

Step 1 — Selecting Your Project | The Most Important Decision You Make

"Do not pick a project that sounds impressive. Pick a project that solves a real problem you have personally seen. The best MVPs come from engineers who were frustrated by something and decided to fix it."

📋 Option A — From Our Project List

We have provided a curated list of 100 AI project ideas across all 6 engineering branches. Each has a defined problem, suggested tech stack, and dataset pointer. Start here if you have no strong idea yet.

Best for: pods that want a proven problem statement with clear boundaries and existing datasets

💡 Option B — Your Own Idea

You have seen a real problem in your domain — in a factory visit, a lab, an internship, or your own campus. You want to build an AI solution for it. Start with the problem, not the technology.

Best for: pods with direct domain access, contacts who experience the problem, or real data sources

🏛 Option C — Ongoing College Project

You already have a project from another subject or a minor project submission. You want to add an AI layer, upgrade the tech stack, and make it deployable and demonstrable.

Best for: pods that want to leverage existing work. You already know the domain — add the AI layer now

🚨 **Do not pick a project because 'AI would be cool in it.' Pick a project because a real person has a real problem that costs them real time or money.**

02

Step 2

Form Your Pod

10 people. One team. One project. One shared commitment until 8th May.

Step 2 — Forming Your Pod | Rules, Roles, and Responsibilities

"A pod is not a study group. It is a product team. Every person has a specific role, a specific deliverable, and a specific accountability. Set this up correctly today and the next 4 weeks will run smoothly."

Pod Formation Rules

Pod size	Exactly 10 members. No pod under 8 or over 12.
Department	Same department or cross-department — your choice.
Pod Leader	One person. Accountable for all milestones. Selected by pod vote today.
Project lock	Your project does not change after Blueprint Lock (end of Day 2).
Weekly check-in	Every pod submits a status update every Friday by 6PM.
Build commitment	Every member commits to a minimum of 8 hours of build work per week.

Suggested Pod Roles

Pod Leader / PM	*Overall accountability. Weekly check-in. Blueprint ownership.*
AI Engineer (×2)	*Claude integration, Flowise RAG, N8N agents.*
Data Engineer	*Azure SQL, dataset, schema, data pipeline.*
Frontend Dev	*Emergent app, UI, user journey.*
DevOps	*GitHub, CI/CD, Antigravity deployment.*
BI / Dashboard	*Power BI, KPI cards, data visualisation.*
Domain Expert (×2)	*Domain knowledge, user research, test validation.*
Demo Lead	*Chronicle HQ site, demo script, presentation.*

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Step 3

Problem Statement

The north star of your entire project. Write it wrong and you build the wrong product.

Step 3 — Problem Statement | The 5-Field Formula

"Everything you build — your architecture, your dataset, your MVP, your pitch — is the answer to your problem statement. It must be specific, measurable, and grounded in a real person's real pain."

WHO	**The specific person or role — not a category**	*e.g. The maintenance engineer on night shift at a chemical plant*
SUFFERS FROM	**The specific measurable pain**	*e.g. spends 4 hours per shift manually checking 23 pressure gauges*
WHICH CAUSES	**Business or safety consequence — in numbers**	*e.g. 2–3 undetected anomalies per week, each costing ₹2–5 lakhs*
WE WILL BUILD	**Your AI solution in one sentence**	*e.g. an AI system that monitors sensors and alerts engineers when action is needed*
SUCCESS MEANS	**Primary KPI — specific, measurable**	*e.g. Detection rate > 95%, response time < 3 minutes, false positive rate*

🚨 **Vague problem = vague product. 'We want to improve efficiency' is NOT a problem statement. Push back every time until every field is specific.**

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Step 4

Build Your AI Co-Founder

A Claude Project that knows your domain, your stack, and your constraints — available 24/7.

Step 4 — AI Co-Founder | Your 24/7 Technical Advisor

"Build a Claude Project with a system prompt that contains everything about your project. This becomes your team's on-demand technical advisor, code reviewer, architecture consultant, and decision support system."

System Prompt — Paste Into Claude Project

You are the AI Co-Founder for [Pod Name].

Project: [Project Name]
Problem: [Your one-line problem statement]
Domain: [Branch + specific area]
User: [Specific person who uses this]
Stack: [All tools selected in Step 5]
Timeline: 4 weeks to working MVP
Team: [N] students, final year [branch]
KPI: [Your primary metric + target number]
Constraints: free tools only, cloud-hosted

Your role: technical advisor, architect,
code reviewer, and decision support.

Test It With These Questions

Before moving to Step 5, ask your AI Co-Founder at least two of these. The answers must be project-specific — not generic.

1. *"What is the most likely technical failure point in our architecture?"*
2. *"Suggest the SQL schema for our primary Azure SQL table with correct data types."*
3. *"What are the 3 most important things to test before our first pod demo?"*
4. *"Write a one-paragraph problem statement for our project in investor language."*

💡 **If the AI Co-Founder answers generically — "it depends on your system" — your system prompt needs more detail. Add more context and re-test.**

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Step 5

Tool Stack + Dataset

Select every layer of your system. Load your first 100 rows. Do both today.

Step 5A — Tool Stack | One Tool Per Layer, All Free, All Covered in This Programme

"You are not choosing tools you like. You are choosing tools that solve your specific problem within your constraints: free, cloud-hosted, and taught in M3–M8."

Layer	Recommended (Use This)	Alternative
Frontend / UI	Emergent (M3)	*Lovable or Bolt — if full redesign needed*
Backend / Hosting	Antigravity (M6)	*Vercel — frontend-only projects*
Database	Azure SQL (M5)	*Supabase — if real-time features needed*
AI / Intelligence	Claude API via Antigravity MCP	*OpenAI — if vision or code interpreter needed*
RAG / Document AI	Flowise (M7)	*NanoNets — if PDF extraction needed*
Automation / Agents	N8N (M8)	*Skip — if MVP is simple query-response*

💡 **Add every tool you select to your AI Co-Founder system prompt now. The more specific the stack, the better the advice you get from it.**

Step 5B — Dataset | No Dataset = No AI | Load 100+ Rows Today

"Every pod decides their data source today and loads it into Azure SQL before leaving. A pod that says 'we will collect data later' does not have a project — they have an idea."

Synthetic — Recommended for MVP

Generate using Claude. Statistically realistic, properly structured, loadable in 10 minutes. Use this if you have no real data access yet.

When: Always valid for MVP. Judges understand synthetic data — what they judge is whether the AI system works.

Public Dataset

Download from Kaggle, UCI ML Repository, data.gov.in, or Google Dataset Search. Clean it, reformat it, load it.

When: Use when your problem matches a well-known domain problem with existing data.

Real Data

Collected from real sensors, forms, logs, or partner organisations. Most credible for judges.

When: Use only if you have confirmed access today. Do not plan for data you might get in Week 3.

SYNTHETIC DATA PROMPTS — Paste into Claude. Branch-specific. Ready in 60 seconds.

Mech	*500 pump maintenance records: date, pump_id, fault_type (Bearing/Seal/Electrical), severity (1-10), downtime_hrs, shift.*
ECE	*180 days signal quality: date, sensor_id, quality_score (0-100), noise_db, calibration_status, location_zone.*
Civil	*200 inspection records: date, site_id, health_index (0-100), load_kn, crack_detected (bool), recommendation.*
Chemical	*300 reactor readings: timestamp, temp_c (85-95 normal, >110 unsafe), pressure_bar (2.0-2.5 normal), status*
CSE/IT	*400 support tickets: timestamp, category, priority, resolution_time_min, error_code, department, resolved (bool).*
EE	*12 months energy: month, department, kwh, peak_demand_kw, cost_inr, efficiency_score (0-100).*

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Step 6

Risk Register

Name the failures before they happen. 5 risks minimum. Written before you start building.

Step 6 — Risk Register | What Will Go Wrong — And How You Will Handle It

"The risk that kills most student projects: 'we assumed the data would be easy to get.' Write your risks today. If you can name the failure mode, you can prevent it."

Data not available by Week 2	HIGH	CRITICAL	*Fix: Decide synthetic vs real today. Load sample data before leaving.*
Pod member stops contributing after Week 1	HIGH	HIGH	*Fix: RACI assigns clear ownership. Weekly check-in submitted every Friday.*
Technical complexity underestimated	HIGH	HIGH	*Fix: MVP scoped to one core feature. Complexity is v2.*
API costs exceed free tier limits	MEDIUM	MEDIUM	*Fix: All chosen tools have free tiers. Azure SQL + Antigravity + Claude all have budget controls.*
Pod cannot agree on direction in Week 2	MEDIUM	HIGH	*Fix: Pod leader has final decision after 10-minute discussion. No exceptions. No re-votes.*
Demo does not work on hackathon day	MEDIUM	CRITICAL	*Fix: Full end-to-end demo tested 3 days before 8th May. Backup screenshots ready.*

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Step 7

Technical Architecture

Draw your 5-layer system before writing a single line of code.

Step 7 — Technical Architecture | The 5-Layer System Model

"Architecture is not a diagram — it is a set of decisions. What data comes in, how AI processes it, where it is stored, what the user sees, and what action results. Draw this on paper first. Then build it in Napkin AI."

Layer 1 DATA LAYER	**Where data enters your system**	*Azure SQL schema, data source (synthetic/real/API), ingestion method, update frequency*
Layer 2 AI LAYER	**What Claude specifically does**	*Input to Claude: [what text/data], Output from Claude: [what response], RAG via Flowise OR direct API call*
Layer 3 LOGIC LAYER	**Automated decisions and rules**	*IF conditions, N8N triggers, agent rules, threshold checks, routing logic*
Layer 4 APPLICATION LAYER	**What the user interacts with**	*Emergent app UI, chatbot interface, API endpoint, admin dashboard*
Layer 5		

💡 **Build your architecture diagram in Napkin AI: paste a text description of your 5 layers → click the lightning bolt → select Architecture Diagram → export as PNG for your GitHub docs/ folder.**

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Step 8

RACI Matrix

Every task has one owner. If everyone is responsible, no one is.

Step 8 — RACI Matrix | Who Does What — No Ambiguity

"R = Does the work (one person). A = Accountable for outcome. C = Consulted before action. I = Informed of outcome. Every task gets exactly one R. No exceptions."

RACI Rules

Every task has exactly one R
One person does it. Not the pod — one person.

Pod Leader is A for all milestones
Accountable does not mean does the work.

No person has R for >40% of tasks
Spread the load. No single points of failure.

If there is no R — it will not get done
Unassigned tasks are invisible tasks.

Revisit RACI every week
Ownership can shift as the project evolves.

Template RACI

Task	M1	M2	M3	M4	M5
Azure SQL + data					
Claude AI layer					
Emergent frontend					
N8N automation					
Power BI dashboard					
Testing + QA					
Demo presentation					
Weekly check-in					

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Step 9

Gantt Chart

4 weeks. Assessment hold respected. Every task has one name against it.

Step 9 — Gantt Chart | Your 4-Week Delivery Plan

"The Gantt chart is not a wishlist. It is a commitment. Open your actual April–May calendar right now. Every task gets a real date and one real person."

WEEK 1 — 7–11 April

- Azure SQL loaded — 100+ rows, schema matches architecture diagram
- Emergent app extended — new UI elements for your specific project flow
- Claude API connected — test response confirms domain-specific output
- Architecture diagram finalised in Napkin AI — PNG saved to GitHub

WEEK 2 — 23–25 April

- Core AI feature working end-to-end — one complete user journey
- N8N automation flow connected to Azure SQL — tested 3 times
- Power BI dashboard live — KPI cards showing real project data
- First internal pod demo — recorded and shared in group

WEEK 3 — 28 Apr – 2 May

- All MVP must-have features complete and individually tested
- Edge cases handled — missing data, wrong input, API timeout
- GitHub CI/CD active — every push auto-deploys to Antigravity
- CV updated — GitHub URL and live demo link added

WEEK 4 — 5–8 May

- 3-minute demo rehearsed minimum 3 times within the pod
- All known bugs fixed — or documented with clear workarounds
- Live URL confirmed from mobile + desktop + not on college WiFi
- 8th May — Hackathon. Every pod presents their live product

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Step 10

MVP Plan + Feature List

Minimum means minimum. One working user journey. Everything else is v2.

Step 10 — MVP Plan | The Feature Filter + Tier System

"For every feature proposed, ask: can you demo your core value without it? If yes — it is not MVP. A great MVP is not a product with fewer features. It is a product that proves exactly one thing, perfectly."

The 3-Question MVP Filter

Apply to EVERY proposed feature. If any answer is YES → not MVP.

Q1 Can you demo your core value without this feature?

→ YES → Remove to v2

Q2 Would judges understand your product without this?

→ YES → Remove to v2

Q3 Does this feature take more than 3 days to build?

→ YES → Simplify or remove

Feature Tier System

MUST HAVE

Without these the product does not work. Maximum 3 features. These are ALL you build for MVP.

SHOULD HAVE

Improves the demo. Build after must-haves work. Maximum 3.

NICE TO HAVE

Only if time allows in Week 4. Do not plan for these.

OUT OF SCOPE

Explicitly excluded. Write these down — stops the pod revisiting them every session.

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Step 11

Definition of Done

One testable sentence per must-have feature. Not 'it works' — a test you can run.

Step 11 — Definition of Done | How You Know Each Feature Is Complete

"Definition of Done removes the most common argument in any project team: 'is this done?' Write the test before you write the code. When the test passes — and only when — the feature is done."

Azure SQL Data Layer	*Done when: SELECT query returns correct rows with no errors. 100+ records loaded. Schema matches architecture diagram exactly.*
Claude AI Integration	*Done when: Given a test input, Claude returns a domain-specific response in under 5 seconds. Tested with 10 different inputs including edge cases.*
Core Application Feature	*Done when: A user completes the primary action from start to finish without error. Works on Chrome, Edge, and mobile browser from a different network.*
N8N Automation Flow	*Done when: Trigger fires when condition is met. Action executes within 30 seconds. Tested 3 consecutive times without failure.*
Power BI Dashboard	*Done when: Dashboard loads in under 3 seconds. All KPI cards show correct values from live Azure SQL data. Shareable URL works without login.*

🚨 **Any task where Done = "we think it works" is not done. Write a specific, runnable test. No test = no done.**

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Step 12

Demo Script

Write your 3-minute hackathon pitch today. If you cannot describe it now, your MVP is wrong.

Step 12 — Demo Script | 3 Minutes. No Slides. Live URL Only.

"The demo script written today does two things: proves you know what you are building, and becomes the practice script for 8th May. Pods that write it today will have rehearsed it 10 times before the hackathon."

0:00–0:30 **THE HOOK**	**Start with the pain. Name the person. Name the cost.** *'Every year, [specific plant/hospital/college] loses ₹[X] because [specific engineer/manager] cannot [specific task]. Nobody has solved this with AI until now.'*
0:30–1:00 **THE DEMO**	**Show the product working. Real data. One complete journey.** *Open the live URL. Do not use screenshots. Enter real data. Show the AI response. Show what the user sees.*
1:00–1:30 **THE RESULT**	**Show the KPI. Before vs After. One number the judge remembers.** *'Before: [baseline]. After our system: [result]. That is a [X]% improvement. Here is the Power BI dashboard showing this live.'*
1:30–2:30 **THE BUILD**	**How it works. Architecture in one sentence. Tools used.** *'We built this on [tools]. The AI layer uses [Claude/Flowise] to [specific function]. The data flows from [source] to [Azure SQL] to [dashboard].'*
2:30–3:00 **THE ASK**	**Specific next step. Not money — a partner, a user, a pilot.** *'We need [a real plant to pilot this / 20 users to validate / an industry partner]. If you can connect us — here is our contact.'*

Blueprint Lock — The 12-Point Sign-Off Checklist

"Every pod confirms all 12 deliverables are complete before leaving. The Blueprint is not a formality — it is your team's commitment to the build ahead. Sign it and mean it."

01 ☐ **Problem Statement**
5-field formula complete, specific, measurable

02 ☐ **AI Co-Founder**
Claude Project built, tested with 2+ domain questions

03 ☐ **Tool Stack**
All layers decided, written down, justified

04 ☐ **Dataset**
100+ rows in Azure SQL, schema matches architecture

05 ☐ **Risk Register**
5+ risks with probability, impact, mitigation

06 ☐ **Architecture**
5-layer diagram in Napkin AI, PNG saved to GitHub

07 ☐ **RACI Matrix**
Every task has one R, pod leader A for all milestones

08 ☐ **Gantt Chart**
4-week plan, hold respected, every task has one owner

09 ☐ **MVP Feature List**
Must Have / Should Have / Out of Scope all filled

10 ☐ **Definition of Done**
Testable sentence per must-have feature

11 ☐ **Demo Script**
3-minute structure written, hook through ask

12 ☐ **GitHub Repo**
Created, README updated, cloud environment active

Weekly Check-In Template | Every Pod — Every Friday by 6PM

"From the Friday after this workshop, every pod submits this every week. 8 fields. 10 minutes to fill. This is how we support all pods without being in the room."

Pod Name + Project	*Pod writes here*
Week Number	*e.g. Week 1 of 4*
What we completed this week	*List each task with the person who did it*
What we did NOT complete — and why	*Be specific. Most important field.*
Current blockers	*Technical, data, team — all count*
What we build next week	*Specific tasks, owners, dates*
Do you need instructor support?	*Yes — describe what / No*

💡 **A pod with momentum score 1 or 2 for two consecutive weeks gets a direct call — not a WhatsApp message. Intervene before Week 3 or it is too late to save the project.**

The Hackathon Pathway — From Blueprint to Stage

"Build well for 4 weeks. Present your live product on 8th May. The top 10 go in front of industry judges on 9th May. The top 3 get rewarded and may get the opportunity to take their product to market."

9–11 Apr

Build Sprints

Three online sessions. Pod-wise support. Every blocker resolved. Every pod building simultaneously.

24 Apr

Placement Prep

Voice assessment. Bot interview simulation. AI CV building. HackWithInfy mock test.

4 May

Final Support

MVP polish. CV finalisation. Demo rehearsal. Every pod tested end-to-end.

8 May

Hackathon

Every pod presents their live product. All 5 layers running. Top 10 selected by judge panel.

9 May

Grand Finale

Top 10 present to industry judges. Top 3 projects rewarded. Certification + prize distribution.

Top 3 projects selected by industry judges may be commercialised — with AIForEveryone.ai support, HIT incubation, and joint IP agreement.

What This Means at Your Placement Interview

"Every recruiter at Infosys, TCS, Wipro, and every startup will ask one question: show me something you built. Not describe — show. A student with a live URL is in a different category from every other candidate in the room."

JD: Infosys Power Programmer — full-stack AI applications, cloud platforms, DevOps	**You: Emergent app + Claude API + Azure SQL + Antigravity deployment + GitHub CI/CD. All connected. All live.**
JD: AI & Agentic Engineering — cloud AI platforms, open source	**You: N8N agent monitoring Azure SQL + Claude analysis + automated WhatsApp alerts. Supervised and unsupervised demonstrated.**
JD: PoCs and MVPs — demonstrable, production-grade	**You: Live URL. Real data. Real AI responses. 3-minute demo that a judge can interact with — not a PowerPoint.**
JD: Bot interview — explain how AI systems work	**You: You can explain embeddings, RAG, agents, and CI/CD with a live demo. Not theory — something you built.**
JD: Open source contributions — GitHub visibility	**You: Public GitHub repo with real commits, CI/CD pipeline, proper project structure. Recruiters can see the history.**

Build it.

Deploy it.
Own it.

You have the tools. You have the plan.
You have 4 weeks. The only variable is execution.

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```
// Blueprint Lock
const blueprint = {
  problemStatement:  '✓',
  aiCoFounder:       '✓',
  toolStack:         '✓',
  dataset:           '✓',
  riskRegister:      '✓',
  architecture:      '✓',
  raci:              '✓',
  gantt:             '✓',
  mvpPlan:           '✓',
  definitionOfDone:  '✓',
  demoScript:        '✓',
  githubRepo:        '✓'
};
// Now go build.
```